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Enhancing GNSS Spoofing Detection by Carrier-Smoothing Pseudorange Double Difference Measurements

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Abstract The presence of counterfeit GNSS signals (commonly referred to as spoofing), is an increasing threat to the integrity of modern navigation systems. These spoofing attacks can fool receivers into processing false signals, resulting in inaccurate position and time data. A simple yet effective method to detect spoofing is by using two separate receivers and computing the double difference (DD) of carrier phase measurements. The DD provides angle of arrival (AOA) information, which helps classify whether a satellite signal is genuine or spoofing. Building on this idea, we propose an improved dual-antenna GNSS spoofing detection technique. Instead of using carrier phase measurements to compute the DD, our approach relies on carrier-smoothed code pseudoranges. This eliminates the need to resolve integer ambiguities inherent in carrier phase-based DD calculations. Experimental results on a mixed tracking dataset consisting of authentic and spoofing satellites demonstrate the effectiveness of our proposed method.

Keywords: Spoofing detection, AOA, Double difference, Smoothing pseudorange.

1. Introduction

Nowadays, the Global Navigation Satellite System (GNSS) has become a more important service that is widely used for civilian and military purposes. Several fields, such as automotive and drone applications, require a highly accurate and authentic GNSS signal. However, the GNSS signal is vulnerable to interference because it is transmitted in a wireless channel with very low power. In

addition, some GNSS signal structures are public, and many hackers have been trying to make a fake GNSS signal to mislead the position and time information of the victim receivers. This kind of attack is also known as ‘spoofing’, one of the most dangerous attacks. Finding a way to determine whether the received signal is authentic or spoofed is therefore mandatory.

Many studies have proposed solutions to address the problem of detecting spoof attacks. Detection techniques can be classified into three levels: signal level, measurement level, and position level [1] [2]. At the signal level, adding cryptographic signatures to navigation messages to verify authenticity is the most effective way. For instance, QZSS recently introduced an authentication service called CLAS (Centimeter Level Augmentation Service) and MADOCA (Multi-GNSS Advanced Orbit and Clock Augmentation), which enhances signal integrity. Furthermore, Japan has implemented the QZSS Authentication Service (QAS) to verify the authenticity of the signal [3]. Margaria et al. provided an in-depth review of civil GNSS authentication schemes, analyzing their robustness against various spoofing techniques and their implementation feasibility [4]. This method, however, requires GNSS receivers’ hardware and/or firmware to be upgraded with new functions to encode and utilize the authenticated signal.

At the position level, the only way to verify the receiver’s final PVT output is by comparing it with external sources, such as cellular-based services, Wi-Fi, an inertial measurement unit (IMU), or eLORAN (enhanced long-ranged navigation) [5]. The disadvantage of this method is that it requires availability of additional sensors (IMU, cellular, Wi-Fi) which may not always be present. We all see that both signal level and position level have their own limitations. Signal level requires some modification of GNSS signal structure whereas position level relies on alternative sources who also provide position service.

In this work, we will focus on detection techniques at the measurement level, which do not require any additional hardware or modifications to the existing GNSS signal structure. Measurement-level techniques operate directly on observables such as pseudorange, carrier phase, Doppler shift, and signal strength, allowing for spoofing detection without relying on external positioning sources or cryptographic enhancements. This approach ensures compatibility with existing GNSS receivers while providing a cost-effective and scalable solution for detecting spoofing attacks. In this approach, spatial processing based on the angle of arrival (AOA) is known as the most simple but robust technique to detect counterfeit signals [6]. This defense exploits the fact that the counterfeit signals are likely to be transmitted from the same direction whereas genuine GNSS signals come from different angles (signals transmitted by different satellites in different orbits).

In [7] and [8], Borio and Gioia introduced a Sum-of-Squares (SoS) detector for GNSS spoofing detection, leveraging carrier phase measurements from dual receivers to detect anomalies in signal arrival angles. Based on the Generalized Likelihood Ratio Test (GLRT) approach, the SoS detector is

computed for every epoch of the receiver. This detector is simple enough and therefore suitable for real-time implementation. However, it will not work properly in case of mixed tracking conditions where both authentic and fake satellites exist in the signal. Derive from the idea of SoS detector, in [9], Nguyen et al. proposed D3 detector which is the dispersion of double difference measurements. Based on the assumption that the DD values of two fake satellites with the same reference satellite are almost the same, authors compared the DD values for every combination of three satellites to find the counterfeit satellites. The D3 method is reported to perform effectively in mixed tracking scenario. The limitation of this method is that the decision threshold is determined empirically based on zero-baseline fractional DD measurements under controlled conditions.

In [10], authors provided an improvement version of D3 method. The decision threshold is set based on a mathematically rigorous approach rather than empirical estimation as that of original D3. In particular, the detection threshold was given according to a target probability of missed detection (P_{mD}), making the method more statistically robust. Although the method provides a theoretical threshold derivation, the standard deviation of noise in DD measurements (used to set the threshold) still needs to be estimated from real data. This prevents the method from fully autonomous operation.

Recently, in [11], the authors utilized a machine learning approach with the Gaussian Mixture Model (GMM) to determine whether a group of four satellites includes any spoofed satellites or not. Using a group of four satellites, the authors selected one satellite as reference. They then calculated three DD values by comparing the three remaining satellites to the reference one. Each satellite can therefore be represented as a point in three-dimensional space based on its computed DD values. The authors claim that the hidden correlation between DD values of real satellites can be learned by using a statistical model, GMM. The method does not require a manual decision threshold but depends on the training data. In addition, the method is limited to classifying a four-satellite group as either containing a spoofed signal or not, but it does not pinpoint which specific satellite within the group is fake. In [12], authors proposed to build a feature vector for every single satellite based on carrier phase DD measurements. The feature vectors are formed by statistical parameters of a series of consecutive DD values computed for each reference satellite. These feature vectors are then fed into a SVM to classify if satellites are authentic or fake ones. Like GMM method, this method also relies on full-period collection of GNSS data with manually labeled spoofed and authentic signals for training phase.

A common point of all the existing works mentioned is their reliance on carrier phase measurements. This raises two questions: (i) Why did they not use the pseudorange measurements? (ii) What is the limitation of using carrier phase measurements? There are two reasons to answer the first question. First, carrier phase measurements offer millimeter-level precision, whereas pseudorange measurements only provide meter-level accuracy. Given that the distance between the two antennas is only a few meters, pseudorange measurements are not suitable for this application. Second, the pseudorange measurements are more affected by multipath errors, atmospheric delay, making them less reliable for

spoofing detection. As illustrated in Fig. 1, the pseudorange double-differencing (DD) values of fake and real satellites are indistinguishable due to excessive noise.

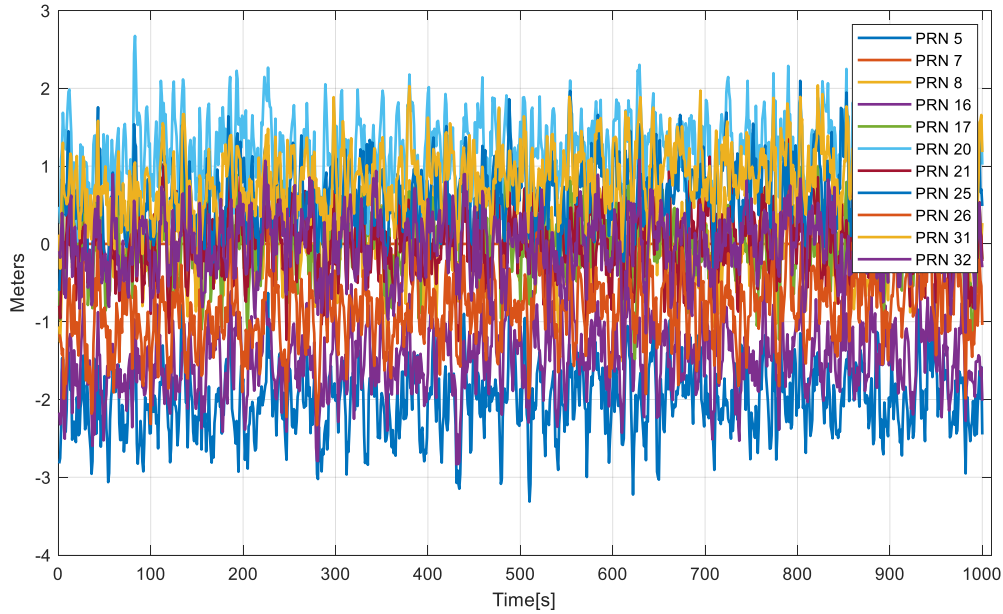


Figure 1. DD values of pseudorange measurements.

For the second question, the limitation of carrier phase double different measurement is the reliance of the integer ambiguities. One solution is to use the *round* operator to round the DD and use only the fractional part of the DD instead of the DD itself, as expressed in following equation:

$fr_{DD} = \Delta^2 \varphi_i - \text{round}(\Delta^2 \varphi_i)$	(1)
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The round operator helps eliminate the impact of the integer ambiguity term in the double-difference (DD) carrier phase. However, this operation can introduce a sign-flipping effect in the fractional DD component. Since the fractional part of the DD carrier phase is constrained within the range $[-0.5, 0.5]$ cycles, applying the round operator may cause abrupt shifts in the remaining fractional value, flipping it from positive to negative and vice versa. As illustrated in Fig. 2, this results in multiple zero-crossing points, which can increase confusion for the spoofing detector.

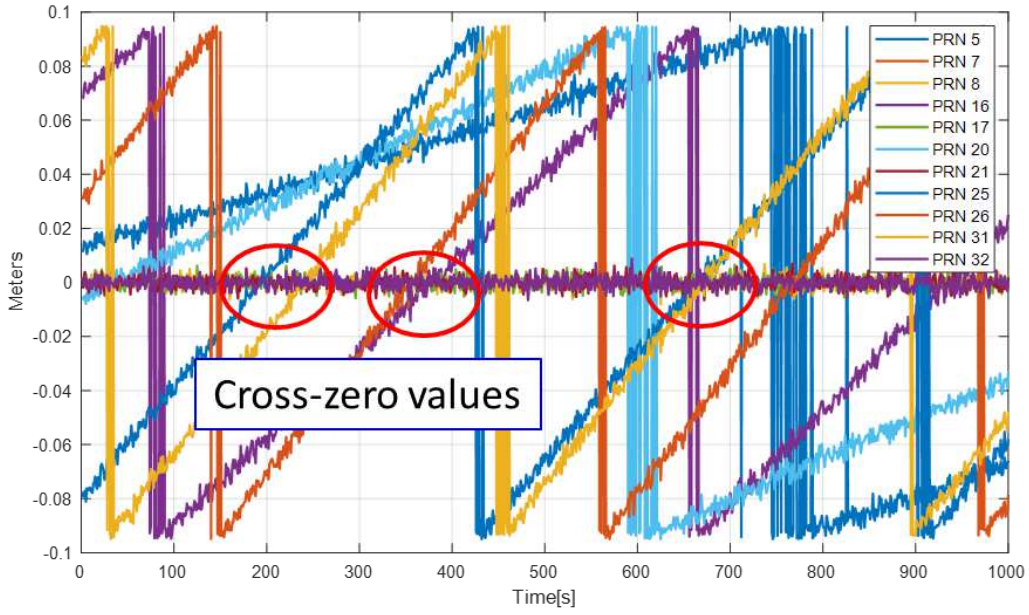


Figure 2. DD values of carrier phase measurements.

In this article, we propose to calculate the DD of not the pseudorange, not the carrier phase but the carrier-phase smoothing pseudorange. Carrier-phase smoothing pseudorange is a technique that applies carrier-phase measurements to refine the pseudorange ones, reducing noise of code and improving accuracy while avoiding integer ambiguity and cycle slips problems associated with the carrier phase observations. The proposed method is evaluated on two distinct datasets:

- Simplistic dataset: the condition of having the whole signal ensemble either counterfeit or authentic.
- Mixed tracking dataset: where both counterfeit and authentic signals are present simultaneously.

For performance assessment, we employ two simple detection techniques: Sum of Squares (SoS) and the D3 detector. Experimental results demonstrate that the proposed double difference of carrier-phase smoothed pseudorange outperforms the conventional carrier-phase double difference method in spoofing detection, particularly in environments where genuine and counterfeit signals coexist. These findings suggest that the proposed approach enhances detection accuracy and robustness, offering a more stable alternative for GNSS spoofing mitigation.

The rest of this paper is organized as follows: Section 2 introduces the signal model and provides a clear comparison between pseudoranges, carrier phase, and carrier smoothing pseudoranges by presenting all relevant models. The derivation of double difference for three measurements is also described in this section. Section 3 briefly describes the SoS and D3 detectors, which will be used to evaluate the proposed carrier-smoothing pseudorange. Many experimental results are shown in Section 4. Conclusions and discussion are provided in Section 5.

2. Signal model

The pseudorange, the carrier phase measurements, and the carrier-smoothed pseudorange for an independent GNSS receiver can be modeled as follows:

$$\rho_i = r_i + c(\delta t_i - \delta T) + \varepsilon_{I,i} + \varepsilon_{T,i} + \varepsilon_i \quad (2)$$

$$\phi_i = r_i + N_i\lambda + c(\delta t_i - \delta T) - \varepsilon_{I,i} + \nu_{T,i} + \nu_i \quad (3)$$

$$\hat{\rho}_i = r_i + c(\delta t_i - \delta T) + \varepsilon_{I,i} + \hat{\varepsilon}_{biasI,i} + \hat{\varepsilon}_{T,i} + \hat{\varepsilon}_i \quad (4)$$

In which,

- Common parameters:
 - i denotes the measurement of the i -th satellite,
 - δt_i is the i -th satellite clock error,
 - δT is the receiver clock error
 - c is the speed of the light,
 - r_i is the geometric range between the receiver and the i -th satellite,
 - $\varepsilon_{I,i}$ is the ionospheric delay
- Parameters for pseudorange:
 - ρ_i is raw pseudorange for the i -th satellite,
 - $\varepsilon_{T,i}$ is the tropospheric error on pseudorange measurement,
 - ε_i is noise term affects to the pseudoranges
- Parameters for carrier phase:
 - ϕ_i is the carrier phase measurement for the i -th satellite by meter,
 - N_i is the integer ambiguous representing the cycle ambiguity,
 - λ is the signal wavelength (19 cm for GPS L1),
 - $\nu_{T,i}$ is the tropospheric errors on carrier phase measurement,
 - ν_i is a noise term accounting for residual unmodeled errors, including thermal noise and multipath.
- Parameters for carrier-smoothed pseudorange
 - $\hat{\rho}_i$ is carrier-smoothed code for the i -th satellite,

- $\hat{\varepsilon}_{biasI,i}$ is the code-carrier divergence effect produced by time varying ionosphere
- $\hat{\varepsilon}_{T,i}$ is the tropospheric errors
- $\hat{\varepsilon}_i$ is noise term after the smoothing filter

The carrier-smoothed pseudorange measurements can be retrieved by configuring the receiver to output these measurements. However, for some low-cost receivers, there is no option to output the carrier-smoothed pseudorange. Fortunately, we can apply the Hatch filter algorithm to smooth the noisy (but unambiguous) code pseudorange measurements with the precise (but ambiguous) carrier phase measurements [13]. The implementation of Hatch filter to compute the smoothed pseudorange is as follows:

$$\hat{\rho}(k) = \frac{1}{n} \rho(k) + \frac{n-1}{n} [\hat{\rho}(k-1) + (\phi(k) - \phi(k-1))] \quad (5)$$

$$\hat{\rho}(1) = \rho(1) \quad (6)$$

$$n = \begin{cases} k, & \text{if } k < N \\ N, & \text{if } k \geq N \end{cases} \quad (7)$$

Where:

- $\hat{\rho}(k)$ is carrier-smoothed pseudorange at the current epoch, k
- $\hat{\rho}(k-1)$ is smoothed pseudorange at the previous epoch, k - 1
- $\rho(k)$ is raw pseudorange measurement at the current epoch, k
- $\phi(k)$ is carrier phase at the current epoch, k
- $\phi(k-1)$ is carrier phase at the previous epoch, k-1
- N is smoothing window width

After filtering, not only is the noisy pseudorange smoothed but the ambiguity integer in carrier phase also canceled. The smoothed pseudorange is modeled as shown in 4, the noise term after the filter smoothing, $\hat{\varepsilon}_i$, is much smaller than the noise term in original pseudorange, ε_i .

Considering two receivers simultaneously observing the same satellites, their output data can be utilized to construct single pseudorange differences, single carrier phase differences, single carrier-smoothed code differences for each satellite in their common view:

$$\Delta\rho_i = \rho_i^{(1)} - \rho_i^{(2)} = (r_i^{(1)} - r_i^{(2)}) + c(\delta T^{(2)} - \delta T^{(1)}) + \Delta\varepsilon_i \quad (8)$$

$$\Delta\phi_i = \phi_i^{(1)} - \phi_i^{(2)} = (r_i^{(1)} - r_i^{(2)}) + \Delta N_i \lambda + c(\delta T^{(2)} - \delta T^{(1)}) + \Delta v_i \quad (9)$$

$$\Delta\hat{\rho}_i = \hat{\rho}_i^{(1)} - \hat{\rho}_i^{(2)} = (r_i^{(1)} - r_i^{(2)}) + c(\delta T^{(2)} - \delta T^{(1)}) + \Delta\hat{\varepsilon}_i \quad (10)$$

Superscripts and ⁽¹⁾ and ⁽²⁾ indicate measurements from the receiver #1 and #2, respectively. For a short baseline between two receivers, ionospheric and tropospheric errors are eliminated. In addition, the

range difference between the satellite and the 7 receivers, denoted as $(r_i^{(1)} - r_i^{(2)})$, can be expressed as shown in [7]:

$$r_i^{(1)} - r_i^{(2)} = D \cos(\alpha_i) \quad (11)$$

where D is the distance between the two antennas and α_i is the angle of arrival (AOA) of the i -th satellite signal. Note that the clock errors are not removed in the single difference measurements. Using double differences, taking the difference between the single difference of i -th satellite and the single difference of ‘ r ’ reference satellite helps eliminate the remaining clock bias term $\delta T^{(2)} - \delta T^{(1)}$ in equations (8), (9), and (10).

$$\Delta \nabla \rho_i = \Delta \rho_i - \Delta \rho_r = D(\cos(\alpha_i) - \cos(\alpha_r)) + \Delta \nabla \varepsilon_i \quad (12)$$

$$\Delta \nabla \phi_i = (\Delta \phi_i - \Delta \phi_r) = D(\cos(\alpha_i) - \cos(\alpha_r)) + \Delta \nabla N_i + \Delta \nabla v_i \quad (13)$$

$$\Delta \nabla \hat{\rho}_i = \Delta \hat{\rho}_i - \Delta \hat{\rho}_r = D(\cos(\alpha_i) - \cos(\alpha_r)) + \Delta \nabla \hat{\varepsilon}_i \quad (14)$$

The term $(\cos(\alpha_i) - \cos(\alpha_r))$ in above DD values depends on the angle of arrival (AOA) of the i -th and the reference received signals. Under normal conditions, GNSS signals are transmitted by different satellites and reach the receiver from various directions, resulting in $\cos(\alpha_i) \neq \cos(\alpha_r)$. In contrast, if counterfeit signals are transmitted from a single source, the received signals will have a common AOA, leading to $\cos(\alpha_i) - \cos(\alpha_r) = 0 \forall i$. Therefore, based on these DD values, spoofing detection can be formulated as the problem of distinguishing between two hypotheses as follows:

$$\begin{aligned} H_0) \quad & D(\cos(\alpha_i) - \cos(\alpha_r)) \approx 0 \quad \forall i \\ H_1) \quad & D(\cos(\alpha_i) - \cos(\alpha_r)) \neq 0 \quad \exists i \end{aligned} \quad (15)$$

where:

- Null hypothesis H_0 indicating that the i -th satellite signal is counterfeit.
- Alternative hypothesis H_1 indicating that the i -th satellite signal is authentic

The term $D(\cos(\alpha_i) - \cos(\alpha_r))$, however, cannot be observed directly. Using pseudoranges in Eq. (12), the DDs are affected by the noise term, $\Delta \nabla \varepsilon_i$, which is even larger than the value of $D(\cos(\alpha_i) - \cos(\alpha_r))$ component. Therefore, the DD of pseudoranges, $\Delta \nabla \rho_i$, does not capture the difference in AOA between two satellites. Instead, it primary reflects the noise measurement. In other words, $\Delta \nabla \rho_i$ is not a reliable spoofing indicator, as it does not provide meaningful information about signal directionality to distinguish between genuine and counterfeit signals. In Eq. (13), we can see that the carrier phase DD values are affected by the ambiguities $\Delta \nabla N_i$ and noise terms $\Delta \nabla v_i$. The noise term, $\Delta \nabla v_i$, in the case of carrier phase is negligible compared to the value of $D(\cos(\alpha_i) - \cos(\alpha_r))$ and can therefore be ignored, we need to remove the ambiguity term though. To remove this component, we can calculate the fractional of the DD as follows:

$$\Delta\nabla\phi_{i,frac} = \Delta\nabla\phi_i - \text{round}(\Delta\nabla\phi_i) \quad (16)$$

Instead of directly using carrier phase DDs, the fractional part of the DDs is now used as the spoofing indicator. In this paper, we propose using the smoothed code DD as spoofing indicator. In Eq. (14), we can see that the DDs of smoothed code are only affected by the noise term $\Delta\nabla\hat{\epsilon}_i$. However, this noisy was smoothed by carrier phase, making it negligible compared to the value of $D(\cos(\alpha_i) - \cos(\alpha_r))$ in case of non-spoofing. The DDs of the smoothed code therefore directly capture the AOA information and can serve as a reliable indicator for spoofing detection.

3. Sum of Squares detector and D3 detector

In this study, we investigate the double difference (DD) of carrier-smoothed pseudorange measurements and demonstrate its effectiveness in spoofing detection, even with simple detection methods. To validate this approach, we employ two existing straightforward detectors: Sum of Squares (SoS) [7][8] and D3 [9]. While the original method was designed to compute SoS and D3 using only the carrier phase DD, we extend its application to pseudorange and carrier-smoothed pseudorange to have a complete comparison among these three measurement types.

3.1. SoS detector

According to [7] and [8], the GLRT approach is proposed to discriminate between H_0 and H_1 at each observation epoch, based on the following test statistic:

$$\Lambda_{SoS}(\mu_i) = \sum_{i=1}^I \omega_i \mu_i^2 \quad (17)$$

Where:

- I is the number of satellites observed in current epoch by both receivers,
- ω_i is a weighted given for each measurement. For simplicity, we set $\omega_i = 1/I$ in this work.
- μ_i is the double difference measurements ($\mu_i = \Delta\nabla\rho_i$ in case of using pseudorange, $\mu_i = \Delta\nabla\phi_{i,frac}$ in case of carrier phase, and $\mu_i = \Delta\nabla\hat{\rho}_i$ in case of smoothed pseudorange measurement).

The value of Λ_{SoS} is then compared with a decision threshold Th . If Λ_{SoS} is greater than Th then the hypothesis H_0 is rejected and normal hypothesis H_1 is declared. The decision threshold is chosen to obtain a target probability of type I error which corresponds to situation when rejecting all null hypothesis even if it is true.

3.2. D3 detector

The D3 detector proposed in [9] did not directly use the DD measurements for spoofing detection but defined a region of similarity for each DD. In particular, the authors formed two test hypotheses for every three satellites as follows:

$$\begin{aligned} H_0) \quad \exists(i, j, k) \in \{S \cup A\}: & \begin{cases} (\mu_i - \mu_k)^2 < \xi_k^2 \\ \text{and} \\ (\mu_j - \mu_k)^2 < \xi_k^2 \end{cases} \\ H_1) \quad \forall(i, j, k) \in \{S \cup A\}: & \begin{cases} (\mu_i - \mu_k)^2 > \xi_k^2 \\ \text{or} \\ (\mu_j - \mu_k)^2 > \xi_k^2 \end{cases} \end{aligned} \quad (18)$$

Where:

- H_0 is null hypothesis means some signals are counterfeit and H_1 is alternative hypothesis where all signals are genuine.
- S is a set of all spoofed satellites and A is a set of all authentic satellites.
- μ_i and μ_k are the double difference measurements of satellite i and k respectively
- ξ_k^2 is a detection threshold which is chosen based on the target probability of missed detection.

For each epoch, the D3 detector representing the difference of DDs between pair of satellites is defined as follows:

$$\Lambda_{D^3}(j, k) = (\mu_i - \mu_k)^2, \quad \forall(j, k) \in \{S \cup A\} \quad (19)$$

4. Experimental Results

4.1. Experimental Setup

To support the theoretical findings described in previous sections, we use the same dataset as in [12], which is a mixed tracking of GPS signals. Figure 3 shows the experimental setup for receiving mixed signals. Three antennas are put outside to receive real signal while the spoofer generates a set of fake GNSS signals that, at the onset of the attack, are time synchronized and perfectly aligned with the ones transmitted by the satellites. This dataset is 700 seconds long including nine genuine satellites (PRNs 2, 5, 12, 14, 16, 15, 25, 26, 29, and 31) and five counterfeit satellites (PRNs 7, 8, 17, 21, and 32). Based on the dataset, we create two new simplistic datasets: one contains only authentic satellites and the other contains all fake satellites. In summary, we have three datasets named as ‘simplistic authentic’, ‘simplistic spoofing’, and ‘mixed tracking’, respectively.



Figure 3. Experimental setup for the mixed tracking dataset [12]

4.2. Experimental Results

4.2.1. Result on simplistic datasets

A. SoS detector results

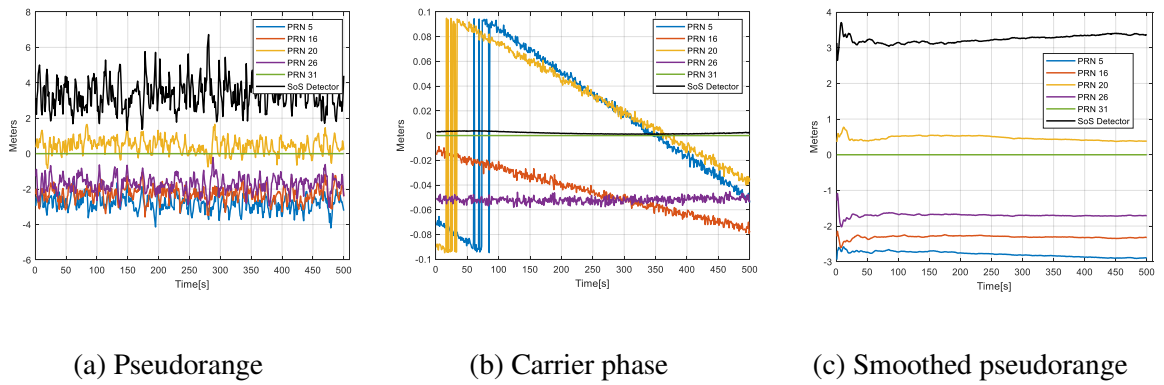


Figure 4. SoS detector results, all satellites are authentic.

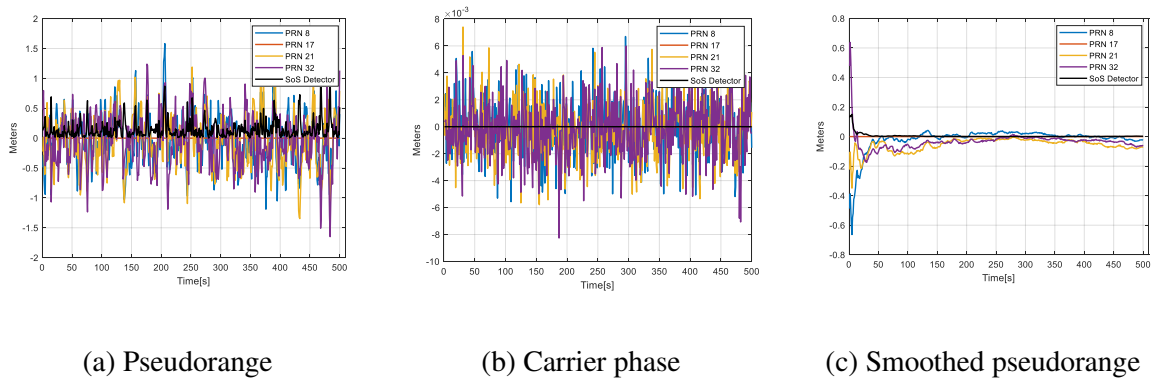


Figure 5. SoS detector results, all satellites are counterfeit.

Figure 4 and Fig. 5 illustrate an example of the SoS detector along with the DD values for three types of measurements (pseudoranges, carrier phases, and smoothed pseudoranges). Figure 4 represents the case where all signals are authentic (simplistic authentic dataset), and Fig. 5 corresponds to the scenario

where all signals are counterfeit (simplistic spoofing dataset). It is intuitively clear that the SoS detector is successfully discriminating between the case where all signals are authentic and the case where all signals are counterfeit. For simplistic datasets, even the pseudorange measurements are good enough for spoofing detection, the Λ_{SoS} in case of counterfeit signals (Fig. 5) is much lower than that of authentic signals (Fig. 4). Moreover, the Λ_{SoS} associated with carrier phase and smoothed pseudorange provide even clearer indications of spoofing detection. Λ_{SoS} is almost zero in the case where all signals are fake.

B. D3 detector results

Unlike the SoS detector, the D3 detector observes the dispersion between the DD measurements. The performance of D3 depends on the relationship of DDs among the three arbitrary satellites. As mentioned in the theoretical section, DDs derived from pseudoranges are significantly affected by noise. Consequently, the D3 detector, Λ_{D^3} , produces false alarms on PRNs 5, 16, and 26, mistakenly identifying these satellites as fake signals despite them being genuine, as shown in Fig. 6 (a). Figure 6 (b) and (c) present the results of the D3 detector for fractional carrier phase DDs and smoothed pseudorange DDs, respectively. As shown in Fig. 6 (b), when using carrier phase, false alarms still occur on PRN 5, 20, and 31 at certain epochs. These false alarms are caused by cycle slips and rounding effects, which make the DDs of these PRNs cross zero, leading to their misidentification as spoofing satellites in those epochs. When using the proposed smoothed pseudorange measurements, the D3 detector performs perfectly in case all satellites are real, as shown in Fig. 6 (c). However, in cases where all satellites are spoofed (simplistic spoofing dataset), the D3 detector initially fails during some of the first epochs, as seen in Fig. 6 (c). This occurs because the smoothed pseudorange is first initialized using the raw pseudorange and is only gradually refined through carrier phase smoothing. As a result, the process requires several epochs to converge, leading to a temporary delay in detecting the spoofing attack. Figures 7 (a) and 7 (b) illustrate the performance of the D3 detector when using pseudorange and carrier phase measurements in a scenario where all satellites are spoofed.

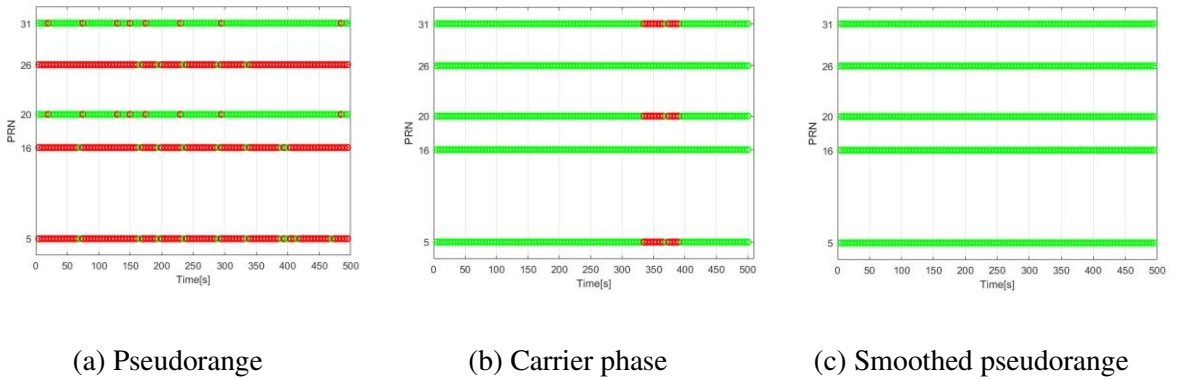


Figure 6. D3 detector results, all satellites are authentic.

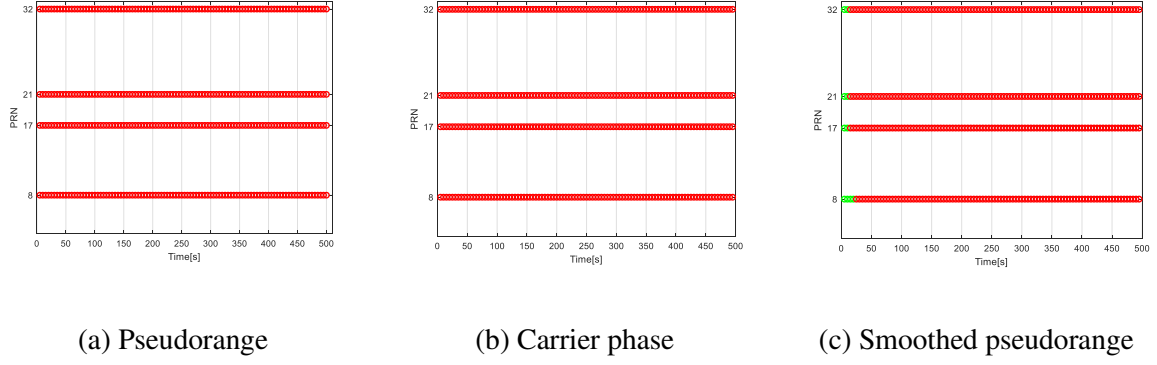


Figure 7. D3 detector results, all satellites are counterfeit.

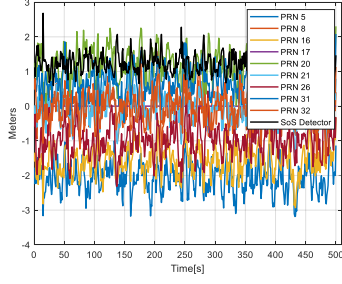
In summary, with the proposed smoothed pseudorange measurements, both the SoS and D3 detectors can distinguish between fully authentic and fully spoofed satellites. Among the three measurement types, double differences (DDs) of raw pseudorange perform the worst, while smoothed pseudorange yields the best results for a simplistic authentic dataset. Table 1 shows the summary results for two simplistic datasets.

Table 1. Correct detection rates for SoS and D3 with Pseudorange, Carrier phase, and Smoothed pseudorange measurements in two scenarios over 500-second dataset.

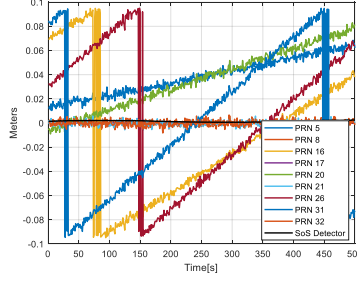
Dataset	Pseudorange				Carrier phase				Smoothed pseudorange			
	SoS		D3		SoS		D3		SoS		D3	
	H_0	H_1	H_0	H_1	H_0	H_1	H_0	H_1	H_0	H_1	H_0	H_1
Simplistic authentic		100 %		42 %		100 %		93.4 %		100 %		100 %
Simplistic spoofing	100 %		100 %		100 %		100 %		96 %		94 %	

4.2.2. Result on mixed tracking dataset

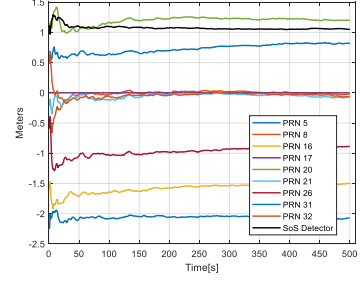
Testing on simplistic authentic and spoofing dataset is obviously not enough, we still need to consider possible scenarios where the victim's receiver locks onto a subset of spoofed satellites while continuing to track the remaining satellites using authentic signals, i.e., mixed tracking dataset.



(a) Pseudorange

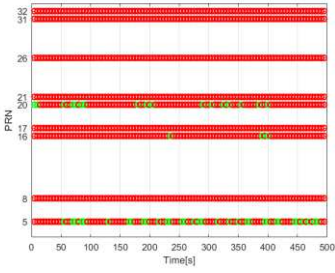


(b) Carrier phase

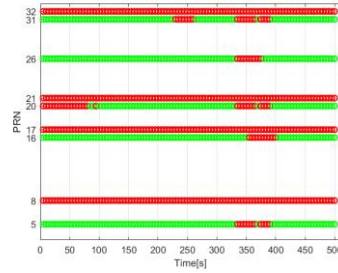


(c) Smoothed pseudorange

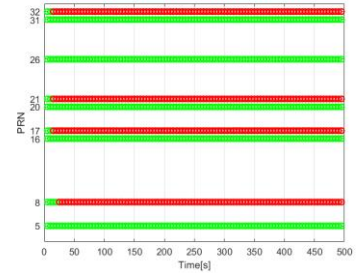
Figure 8. SoS detector results in mixed tracking dataset.



(a) Pseudorange



(b) Carrier phase



(c) Smoothed pseudorange

Figure 9. D3 detector results in mixed tracking dataset.

Figures 8 and 9 show examples of the SoS and D3 detectors, along with the DD values for three types of measurements (raw pseudoranges, carrier phases, and smoothed pseudoranges) over a 500-second mixed tracking dataset, respectively. It can be observed that, for all measurement types, the SoS detector fails to reveal the presence of spoofed satellites, whereas the D3 detector successfully identifies which satellites are real and which are spoofed. To be more specific, the D3 detector fails when using raw pseudorange measurements, achieving 96.6% false alarms when identifying actual satellites, H_1 , as spoofing ones, H_0 hypothesis (as shown in Table 2, columns 4 and 5). When using the double differences (DDs) of the proposed smoothed pseudorange, the D3 detector outperforms carrier phase measurements for the H_1 hypothesis, with a 100% success rate compared to 84.8% for carrier phase measurements (as shown in Table 2). Although the detector's performance with DDs of smoothed pseudorange is slightly lower for the H_0 hypothesis (97.6%) compared to DDs of carrier phase measurements (100%), this does not impact practical implementation. The initial performance gap can be mitigated by skipping the first few epochs to allow the smoothed pseudorange values to stabilize.

Table 3 summarizes the results for the entire dataset, which consists of 7,000 seconds of mixed GPS tracking. The SoS detector fails by classifying all satellites as authentic. In contrast, for the D3 detector, all evaluation metrics indicate that smoothed pseudorange measurements outperform other methods

except for the miss detection metric. However, we can easily reduce the miss detection rate for smoothed pseudorange measurements by skipping the first few epochs.

Table 2. Confusion matrix for 500-second mixed tracking dataset in case of using pseudorange, carrier phase, and smoothed pseudorange measurements.

Predicted \ Actual	Pseudorange				Carrier phase				Smoothed pseudorange			
	SoS (%)		D3 (%)		SoS (%)		D3 (%)		SoS (%)		D3 (%)	
	H_0	H_1	H_0	H_1	H_0	H_1	H_0	H_1	H_0	H_1	H_0	H_1
H_0	0	100	100	0	0	100	100	0	0	100	97.6	2.4
H_1	0	100	96.6	3.4	0	100	15.2	84.8	0	100	0	100

Table 3. Confusion matrix for the whole mixed tracking dataset in case of using pseudorange, carrier phase, and smoothed pseudorange measurements.

Predicted \ Actual	Pseudorange				Carrier phase				Smoothed pseudorange			
	SoS (%)		D3 (%)		SoS (%)		D3 (%)		SoS (%)		D3 (%)	
	H_0	H_1	H_0	H_1	H_0	H_1	H_0	H_1	H_0	H_1	H_0	H_1
H_0	0	100	99.96	0.04	0	100	99.9	1	0	100	93.7	6.3
H_1	0	100	99.67	0.33	0	100	11.6	89.4	0	100	3.8	96.2

5. Conclusion

This paper presented an enhanced GNSS spoofing detection technique utilizing carrier-smoothed pseudorange double difference (DD) measurements. Unlike conventional approaches that rely on carrier phase DD measurements, our method leverages smoothed pseudoranges to mitigate integer ambiguity issues while preserving the angle of arrival (AOA) information necessary for spoofing detection.

Through experimental evaluations on both simplistic and mixed tracking datasets, we demonstrated that the proposed smoothed pseudorange DD approach significantly improves spoofing detection performance. While the Sum-of-Squares (SoS) detector failed in mixed tracking conditions, the D3 detector successfully distinguished between authentic and spoofed satellites. Furthermore, our results showed that using smoothed pseudorange measurements led to higher detection accuracy compared to raw pseudorange or carrier phase measurements, particularly for the H_1 hypothesis.

Although the D3 detector exhibited a slight delay in detecting spoofed satellites due to the initialization process of smoothed pseudorange, this limitation can be mitigated by skipping the first few epochs to allow values to converge. Overall, the proposed technique offers a practical and effective solution for GNSS spoofing detection, maintaining compatibility with existing GNSS receivers without requiring any additional hardware modifications.

Future work could focus on refining the smoothing process to further reduce the initialization delay and integrating machine learning models to enhance detection robustness in real-world scenarios.

6. Data availability

The dataset used in this work is published on GitHub at the following address:

<https://github.com/hiephv/SpoofingDetectionDataset>

The dataset contains two MATLAB files: **rx1.mat** and **rx2.mat**, each corresponding to measurements from a different receiver.

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